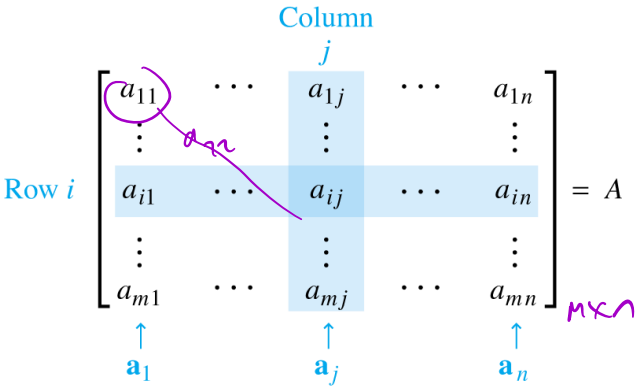
**Linear Algebra Lecture Notes**

## Chapter 2 Matrix Algebra

Basic definitions:

- If $A$ is an $m \times n$ matrix—that is, a matrix with $m$ rows and $n$ columns—then the scalar entry in the $i$th row and $j$th column of $A$ is denoted by $a_{ij}$ and is called the $(i, j)$-entry of $A$. See the figure below.




Matrix notation.

- Each column of $A$ is a list of $m$ real numbers, which identifies a vector in $\mathbb{R}^m$.
- The columns are denoted by $\mathbf{a}_1, \ldots, \mathbf{a}_n$, and the matrix $A$ is written as

$$A = [\boldsymbol{a}_1 \quad \boldsymbol{a}_2 \quad \cdots \quad \boldsymbol{a}_n]$$

  The number $a_{ij}$ is the $i$th entry (from the top) of the $j$th column vector $\mathbf{a}_j$.
- The **diagonal entries** in an $m \times n$ matrix $A = [a_{ij}]$ are $a_{11}, a_{22}, a_{33}, \ldots$, and they form the **main diagonal** of $A$.
- A **diagonal matrix** is a square $n \times n$ matrix whose nondiagonal entries are zero. An example is the $n \times n$ identity matrix, $I_n$.
- An $m \times n$ matrix whose entries are all zero is a **zero matrix** and is written as 0.
- The two matrices are **equal** if they have the same size (*i.e.*, the same number of rows and the same number of columns) and if their corresponding columns are equal, which amounts to saying that their corresponding entries are equal.

- If $A$ and $B$ are $m \times n$ matrices, then the **sum** $A + B$ is the $m \times n$ matrix whose columns are the sums of the corresponding columns in $A$ and $B$.
- Since vector addition of the columns is done entrywise, each entry in $A + B$ is the sum of the corresponding entries in $A$ and $B$.
- The sum $A + B$ is defined only when $A$ and $B$ are the same size.
- If $r$ is a scalar and $A$ is a matrix, then the **scalar multiple** $rA$ is the matrix whose columns are $r$ times the corresponding columns in $A$.

Ex:

$$A = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}_{2\times 3} \qquad B = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 7 & 7 \end{bmatrix}_{2\times 3} \qquad C = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}_{2\times 2}$$

$$A + B = \begin{bmatrix} 5 & 1 & 6 \\ 2 & 8 & 9 \end{bmatrix} \qquad A + C \text{ is undefined.}$$

$$2 \cdot B = \begin{bmatrix} 2 & 2 & 2 \\ 6 & 10 & 14 \end{bmatrix} \qquad A - 2B = \begin{bmatrix} 2 & -2 & 3 \\ -7 & -7 & -12 \end{bmatrix}$$

Theorem: Let $A, B, C$ be matrices of the same size and let $r$ and $s$ be scalars.

a. $A + B = B + A$

b. $(A+B)+C = A+(B+C)$

c. $A + 0 = A$

d. $r(A+B) = rA + rB$

e. $(r+s)A = rA + sA$

f. $r(sA) = (r \cdot s)A$

Remember the definition given in section 1.4:

If $A$ is an $m \times n$ matrix, with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$, and if $\mathbf{x}$ is in $\mathbb{R}^n$, then the **product of $A$ and $\mathbf{x}$**, denoted by $A\mathbf{x}$, is **the linear combination of the columns of $A$ using the corresponding entries in $\mathbf{x}$ as weights**; that is,

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

($m \times n$ times $n \times 1$)

Ex:

$$\underbrace{\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix}}_{x} = 4\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 2 \\ -5 \end{bmatrix} + 7\begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

($a_1$, $a_2$, $a_3$; $x_1$, $x_2$, $x_3$)

$$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 6 \\ -15 \end{bmatrix} + \begin{bmatrix} -7 \\ 21 \end{bmatrix} = \boxed{\begin{bmatrix} 3 \\ 6 \end{bmatrix} = Ax}$$

**Definition:**

If $A$ is an $m \times n$ matrix, and if $B$ is an $n \times p$ matrix with columns $\mathbf{b}_1, \ldots, \mathbf{b}_p$, then the product $AB$ is the $m \times p$ matrix whose columns are $A\mathbf{b}_1, \ldots, A\mathbf{b}_p$. That is,

$$AB = A\begin{bmatrix}\mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p\end{bmatrix} = \begin{bmatrix}A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p\end{bmatrix}$$

Ex: Compute $A \cdot B$ where $A = \begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}$ ($2\times 2$), $B = \begin{bmatrix}4 & 3 & 6\\ 1 & -2 & 3\end{bmatrix}$ ($2\times 3$)

write $B = [b_1\ b_2\ b_3]$ and compute

$Ab_1 = \begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}4\\ 1\end{bmatrix} = 4\begin{bmatrix}2\\ 1\end{bmatrix} + 1\begin{bmatrix}3\\ -5\end{bmatrix} = \begin{bmatrix}11\\ -1\end{bmatrix} \Rightarrow Ab_1$

$Ab_2 = \begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}3\\ -2\end{bmatrix} = 3\begin{bmatrix}2\\ 1\end{bmatrix} - 2\begin{bmatrix}3\\ -5\end{bmatrix} = \begin{bmatrix}0\\ 13\end{bmatrix} \Rightarrow Ab_2$

$Ab_3 = \begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}\begin{bmatrix}6\\ 3\end{bmatrix} = 6\begin{bmatrix}2\\ 1\end{bmatrix} + 3\begin{bmatrix}3\\ -5\end{bmatrix} = \begin{bmatrix}21\\ -9\end{bmatrix} \Rightarrow Ab_3$

Then $A \cdot B = A[b_1\ b_2\ b_3] = [Ab_1\ Ab_2\ Ab_3] = \begin{bmatrix}11 & 0 & 21\\ -1 & 13 & -9\end{bmatrix} = A \cdot B$

Obviously, number of columns of $A$ must match the number of rows of $B$.

Ex: $A$ is $3\times 5$, $B$ is $5\times 2$ then size of $AB$ is $3\times 2$.

what about $BA$? → it is undefined.

- Each column of $AB$ is a linear combination of the columns of $A$ using weights from the corresponding columns of $B$.
- Obviously, the number of columns of $A$ must match the number of rows in $B$. Also, the definition of $AB$ shows that $AB$ has the same number of rows as $A$ and the same number of columns as $B$.

Row-Column Rule for Computing $AB$:

If the product $AB$ is defined, then the entry in row $i$ and column $j$ of $AB$ is the sum of the products of corresponding entries from row $i$ of $A$ and column $j$ of $B$. If $(AB)_{ij}$ denotes the $(i, j)$-entry in $AB$, and if $A$ is an $m \times n$ matrix, then

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Ex:

$A = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}$

2x2 2x3

To find entry in row 1 and column 1 of AB

$A \cdot B = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2\cdot 4 + 3\cdot 1 & & \\ & & \end{bmatrix}$

$\begin{bmatrix} 11 & & \\ & & -9 \end{bmatrix}$

To find entry in row 2 and column 3 of AB

$1\cdot 6 + -1\cdot 3 = -9$

Ex: $A = \begin{bmatrix} 2 & -5 & 0 \\ -1 & 3 & -4 \\ 6 & -8 & -7 \\ -1 & 0 & 9 \end{bmatrix}$ $B = \begin{bmatrix} 4 & -6 \\ 1 & 1 \\ 3 & 2 \end{bmatrix}$

4x3 3x2

find second row of AB.

$-1\cdot 4 + 3\cdot 1 + -4\cdot 3 = -5$

$-1\cdot -6 + 3\cdot 1 + -4\cdot 2 = 1$

$\begin{bmatrix} & \\ -5 & 1 \\ & \\ & \end{bmatrix} = AB$

4x2

row = 2

col = 2

## Properties of matrix multiplication

**Theorem:** Let $A$ be an $m \times n$ matrix and let $B$ and $C$ have sizes for which the indicated sums and products are defined.

a. $A(BC) = (AB)C$ (associative law of multiplication)

b. $A(B+C) = AB + AC$ (left distr. law)

c. $(B+C)A = BA + CA$ (right dist. law)

d. $r(AB) = (rA)B = A(rB)$ for any scalar $r$.

e. $I_m A = A = A I_n$ (identity for matrix multiplication) $I_m, I_n$ is the identity matrix.

**Ex:** Let $A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$ Show that $AB \neq BA$

$$AB = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 5\cdot 2 + 1\cdot 4 & 5\cdot 0 + 1\cdot 3 \\ 3\cdot 2 + -2\cdot 4 & 3\cdot 0 + -2\cdot 3 \end{bmatrix} = \begin{bmatrix} 14 & 3 \\ -2 & -6 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 2\cdot 5 + 0\cdot 3 & 2\cdot 1 + 0\cdot -2 \\ 4\cdot 5 + 3\cdot 3 & 4\cdot 1 + 3\cdot -2 \end{bmatrix} = \begin{bmatrix} 10 & 2 \\ 29 & -2 \end{bmatrix}$$

Observation: $AB \neq BA$

- In general $AB \neq BA$.
- If $AB = AC$ then it is not true in general that $B = C$.
- If a product $AB$ is the zero matrix, we can not conclude that either $A = 0$ or $B = 0$.

## Powers of a matrix

If $A$ is an $n \times n$ matrix and if $k$ is a positive integer, then $A^k$ denotes the product of $k$ copies of $A$.

That is, $A^k = \underbrace{A \cdots A}_{k}$

## Transpose of a matrix

Given an $m \times n$ matrix $A$, the transpose of $A$ is the $n \times m$ matrix, denoted by $A^T$, whose columns are formed from the corresponding rows of $A$.

Ex: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $B = \begin{bmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{bmatrix}_{3\times 2}$

$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$ $B^T = \begin{bmatrix} -5 & 1 & 0 \\ 2 & -3 & 4 \end{bmatrix}_{2\times 3}$

Theorem: Let $A$ and $B$ denote matrices whose sizes are appropriate for the following sums and products.

a. $(A^T)^T = A$

b. $(A+B)^T = A^T + B^T$

c. For any scalar $r$, $(rA)^T = rA^T$

d. $(AB)^T = B^T \cdot A^T$